

CLIENT SIDE TECHNOLOGIES

JavaScript Built-in Objects

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JavaScript Objects

JavaScript objects fall into 4 categories:

1

Custom Objects

Objects that you, as a JavaScript developer, create and use.

2

Built-in Objects

Objects provided with JavaScript to make your life easier.

3

BOM Objects

A collection of objects accessible through the global window object. Deals with browser characteristics and properties.

4

DOM Objects

Provide the foundation for dynamic web pages — access, manipulate, and extend page content.

JavaScript Built-in Objects



String



Number



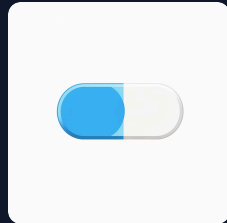
Array



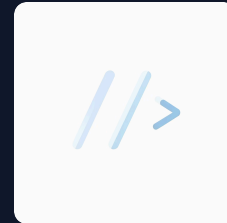
Date



Math



Boolean



RegExp



Error



Object

String Object

Enables us to work with and manipulate strings of text.

One Property

length – gives the length of the String.

Method Category 1

Manipulate the contents of the String

Method Category 2

Manipulate the appearance of the String

Method Category 3

Convert the String into an HTML element

String Object – Content Methods

```
var myStr = "Let's see what happens"
```

Method	Description	Example
charAt(index)	Returns the character at the specified index	myStr.charAt(0) → "L"
indexOf(string)	Returns position of first occurrence	myStr.indexOf("at") → 12 myStr.indexOf("@") → -1
lastIndexOf(string)	Returns position of last occurrence	myStr.lastIndexOf("a") → 16
substring(i,j)	Extracts characters between two indices	myStr.substring(1,7) → "et's s" myStr.substring(2) → "t's see what happens!"
replace(pattern)	Searches and replaces matched text	myStr.replace(/e/,"?") → "L?t's see what happens!" myStr.replace(/e/g,"?") → "L?t's s?? what happens!"

String Object – Appearance Methods

Reference: [w3schools.com/jsref/jsref_obj_string.asp](https://www.w3schools.com/jsref/jsref_obj_string.asp)

bold()

Wraps the string in tags to display bold text.

italic()

Wraps the string in <i> tags to display italic text.

fontcolor(color)

Wraps the string in tags to change text color.

fontsize(size)

Wraps the string in tags to change text size.

link(url)

Wraps the string in an anchor tag to create a clickable hyperlink.

small()

Wraps the string in <small> tags to display smaller text.

Math Object

Allows you to perform common mathematical tasks.

The Math object is a static object — no need to create an instance.

Properties

Math object has:

- Properties (constant values)
- Methods

Methods

Common mathematical tasks are performed using Math methods.

Examples include rounding, random values, and more.

Properties

Math.E

Returns Euler's constant

Math.PI

Returns the value of π (PI)

Example

```
;var circleArea = Math.PI * radius * radius
```

Math Object – Methods

Common JavaScript Math methods at a glance

abs()

Returns the absolute value.

Example:

```
Math.abs(-7) // 7
```

ceil()

Rounds up to the nearest integer.

Example:

```
Math.ceil(4.2) // 5
```

floor()

Rounds down to the nearest integer.

Example:

```
Math.floor(4.8) // 4
```

round()

Rounds to the nearest integer.

Example:

```
Math.round(4.5) // 5
```

max()

Returns the largest value.

Example:

```
Math.max(3, 9, 5) //  
9
```

min()

Returns the smallest value.

Example:

```
Math.min(3, 9, 5) //  
3
```

pow()

Raises a number to a power.

Example:

```
Math.pow(2, 3) // 8
```

sqrt()

Returns the square root.

Example:

```
Math.sqrt(16) // 4
```

random()

Returns a random decimal between 0 and 1.

Example:

```
Math.random()
```

Reference: [w3schools.com/jsref/jsref_obj_math.asp](https://www.w3schools.com/jsref/jsref_obj_math.asp)

Array Object

Create and access arrays in JavaScript

To declare an array:

```
;()var colorArray = new Array
    ;"colorArray[0] = "red
    ;"colorArray[1] = "blue
    ;"colorArray[2] = "green

// OR
;var colorArray = new Array("red", "blue", "green")

// OR
;[]=var colorArray
;var colorArray = ["red", "blue", "green"]
```

Array Object has One Property:

length – gives the length of the array



Indexed elements are stored starting at **0**, so the **length** property counts the total number of items.

Array Object – Methods

Common JavaScript array methods at a glance.

join()

Combines elements into a string.

```
arr.join(",")
```

reverse()

Reverses the array order.

```
arr.reverse()
```

sort()

Sorts items in place.

```
arr.sort()
```

concat()

Joins arrays into a new array.

```
arr1.concat(arr2)
```

slice()

Returns a portion of the array.

```
arr.slice(1, 3)
```

splice()

Adds, removes, or replaces items.

```
arr.splice(1, 2, "X")
```

push()

Adds items to the end.

```
arr.push("D")
```

pop()

Removes the last item.

```
arr.pop()
```

shift()

Removes the first item.

```
arr.shift()
```

unshift()

Adds items to the beginning.

```
arr.unshift("A")
```

find()

Returns the first matching element.

```
arr.find(x => x ===  
"B")
```

filter()

Returns all matching elements.

```
arr.filter(x => x > 10)
```

Array Object – Associative Arrays

The Arrays That Aren't

1

Numeric Arrays

Use numeric indices to access values.

0 → "Ahmed"

1 → "Tareq"

2 → "Sara"

1

Associative Arrays

Use string keys to access values.

"Ahmed" → "Excellent"

"Tareq" → "pass"

"Sara" → "Outstanding"

- Associative arrays let you specify key-value pairs.
- Instead of numeric indices, they use string keys — easier to remember and reference.
- Keys must be strings, but values can be any data type, including other arrays.

Syntax:

```
;()var assocArray = new Array  
;"assocArray["Ahmed"] = "Excellent  
; "assocArray["Tareq"] = "pass
```

Literal Object

What is a Property?

An object is an unordered list of primitive data types stored as name-value pairs.

Properties vs Methods

Each item is called a property; functions are called methods.

What is an Object?

Objects store related data as properties and can also contain methods for behavior.

Code Examples

PROPERTIES

```
;var emp1 = { Name: "Richard", Age: 30 }  
;var emp2 = { Name: "Ali", Age: 20 }  
  
;emp1.Name = "Mahmoud"  
;alert(emp1.Name)
```

METHODS

```
,var emp1 = { name: "Aly"  
  ,age: 23  
  }() show: function  
  alert(this.name + " is " + this.age  
  ;{  
  
;()emp1.show
```

Date Object

To obtain and manipulate the date and time in a script.

JAVASCRIPT

DATE

Syntax

```
;()var d = new Date
```

```
;var d = new Date(dateString)
```

```
;var d = new Date(year, month, day, hours, min, sec, ms)
```

Number Conventions

Date Attribute	Numeric Range
seconds, minutes	0 – 59
hours	0 – 23
day	0–6 (0 = Sunday, 1 = Monday...)
date	1 – 31
month	0–11 (0 = January, 1 = February...)
year	Full 4-digit year (e.g. 2006)

Date Object – Methods

Date methods fall into three categories: get, set, and to methods.

```
var myDate = new Date("November 25, 2006 11:13:55");
```

get Methods

- getDate() — day of the month
- getDay() — day of the week
- getMonth() — month number
- getFullYear() — four-digit year
- getHours(), getMinutes(), getSeconds()

set Methods

- setDate() — set day of the month
- setMonth() — set month
- setFullYear() — set year
- setHours(), setMinutes(), setSeconds()
- setTime() — set time in milliseconds

to Methods

- toString() — convert date to text
- toDateString() — date portion only
- toTimeString() — time portion only
- toISOString() — ISO format
- toLocaleDateString() — localized date

Reference: [w3schools.com/jsref/jsref_obj_date.asp](https://www.w3schools.com/jsref/jsref_obj_date.asp)

Regular Expression Object

Regular expressions provide a powerful way to search and manipulate text.

Pattern Search

A Regular Expression represents a pattern you are looking for in a string.

Special Characters

A RegExp lets you build patterns using special characters; appropriate action is taken on match.

Validation

Often used for validation — you know the format users must enter, not the exact values.

Text Manipulation

Regular expressions provide a powerful way to search and manipulate text.

Syntax

Using RegExp constructor

```
;var searchPattern = new RegExp("j.*t", "i")
```

Using literal syntax

```
;var myRegExp = /j.*t/i
```

Flags

i

Ignore case

g

Global search

m

Multiline

Regular Expressions – Syntax Reference

Char	Description	Example
.	Any character	/a.*a/ matches "aa", "aba", "a9qa"
^	Start	/^a/ matches "apple", "abcde"
\$	End	/z\$/ matches "abcz", "az"
	Or	/abc def g/ matches "abc", "def", or "g"
[]	Match any one char in brackets	/[a-z]/ matches any lowercase letter
[^]	Match any char NOT in brackets	/[^abcd]/ matches any char but a, b, c, d
*	0 or more	/Goo*gle/ → "Gogle", "Google", "Gooogle"
+	1 or more	/Goo+gle/ → "Google", "Gooogle"
?	0 or 1	/Goo?gle/ → "Gogle", "Google"
{n}	Exactly n	/a{3}/ matches "aaa"
\d	Any digit	/\d+/ matches "123"
()	Group	/(abc){2}/ matches "abcabc"

Regular Expressions – Methods

RegExp Methods:

test(str)

Returns **true** or **false** — **true** when there's a match.

```
var reg = /j.*t/;  
reg.test("Javascript"); // false
```

exec(str)

Returns the **first matched string**.

```
var reg = /j.*t/i;  
var str = "Jscript is the same of javascript";  
reg.exec(str); // "Jscript"
```

String Methods

String methods that accept RegExp:

match(), **search()**, **replace()**, **split()**

Use these when you want to check, find, replace, or break apart text using patterns.

Error Object

Whenever an error occurs, an instance of the Error object is created to describe it.

Created in 2 ways

Type	Example
Explicitly	<pre>;()var newErrorObj = new Error</pre>
Implicitly	<pre>.thrown using the throw statement</pre>

Error Object Properties

Property	Description
description	Plain-language description of error
fileName	URI of the file containing the script throwing the error
lineNumber	Source code line number of error
message	Plain-language description of error (ECMA)
name	Error type (ECMA)
number	Microsoft proprietary error number

Error Object – Error Types

Six additional Error constructors exist, all inheriting from Error:

Use `instanceof` when catching errors to identify their built-in type.

EvalError

Error in the global `eval()` function

SyntaxError

Code has invalid syntax or structure

RangeError

A value is not within the expected range

TypeError

A value is not of the expected type

ReferenceError

An undeclared variable or invalid reference was used

URIError

URI handling or encoding failed

```
;var e = new EvalError("jaavcsritp is _not_ how you spell it")
document.write(e.name); // EvalError
```

JavaScript Built-in Objects

Thank You!

```
;document.writeln("Thank You!")
```

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